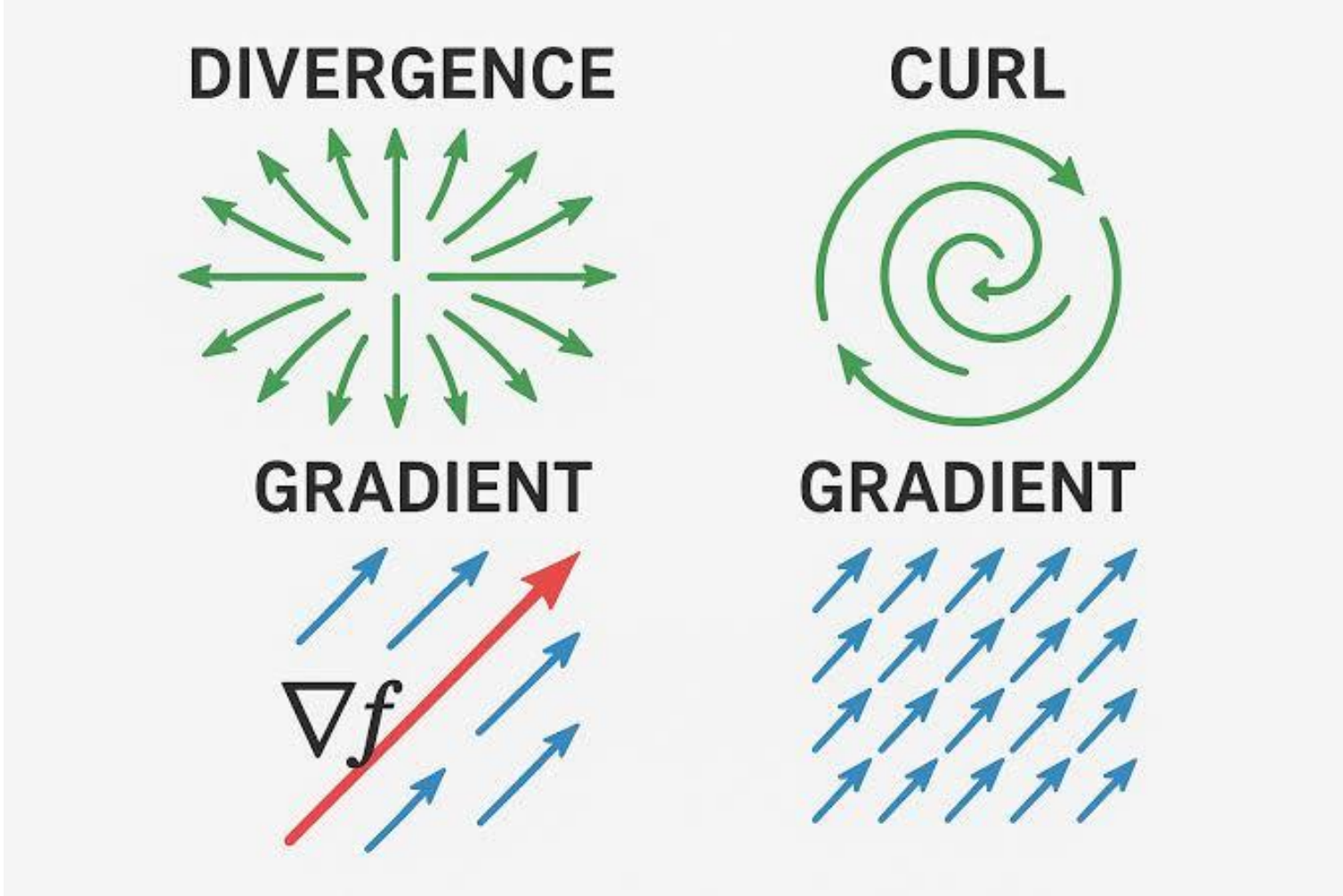
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Gradient, Divergent and Curl



Gradient Vector

Suppose **temperature** in a room is described by $T(x, y) = 20 + x^2 + 2y^2$. At every point (x, y), there is a temperature.

The natural question is:

In which direction does the temperature increase most rapidly ?

This is where the **gradient vector** enters.

Gradient Vector

Imagine standing on a hill.

Suppose the height of the ground is described by $z = f(x, y)$. At your current position, there are many possible directions you could walk. **The gradient** vector tells us the particular direction in which the height z will increase as fast as possible.

The **gradient** takes a **scalar field** and turns it into **a vector field**

Input : Scalar field (f)

Output : Vector field (∇f)

$$\nabla f(x,y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = f_x \mathbf{i} + f_y \mathbf{j}$$

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$$

Gradient Vector

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Example 01

Suppose the temperature distribution over a metal plate is $\boxed{T(x, y) = 100 + 4x + 3y}$ where T is measured in °C and x, y are measured in metres.

Then $\frac{\partial T}{\partial x} = 4, \frac{\partial T}{\partial y} = 3.$

Therefore, $\boxed{\nabla T = 4i + 3j}$ is the direction of greatest temperature increase is the direction of $4i + 3j$.

Example 02

For $f(x, y) = x^2 + 3xy + y^2$, we get $f_x = 2x + 3y, f_y = 3x + 2y.$

Therefore, $\nabla f = \langle (2x + 3y), (3x + 2y) \rangle = (2x + 3y)i + (3x + 2y)j$

At $(1, 2)$, $\nabla f(1, 2) = \langle 8, 7 \rangle$.

Directional Derivatives

Once you **understand** the gradient, then you may ask yourself "What if I don't want to move in the direction of maximum increase? What if I want to move in a specific direction ?"

Directional derivative : A directional derivative measures the rate of change of a multivariable function along a specific direction or vector from a given point.

The **directional derivative** generalizes the standard partial derivative. Instead of moving only parallel to the coordinate axes (like x or y), you move **along an arbitrary direction**.

To **calculate the directional derivative** of a function f **at a point** in the direction of a unit vector $\bar{u}$ **,** you take the **dot product** of the function's gradient (∇f) and the unit vector $\bar{u}$.

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$$

Example Let $f(x, y) = x^2 + y^2$. At $(1, 2)$, $\nabla f(1, 2) = \langle 2, 4 \rangle$. Suppose we want the rate of change toward $v = \langle 3, 4 \rangle$.

OR

Given $f(x, y) = x^2 + 3xy$, find $D_{\mathbf{u}} f(1,2)$in the direction of $\mathbf{v} = \langle 3,4 \rangle$.

$$D_{\mathbf{u}} f(x, y) = \nabla f(x, y) \cdot \mathbf{u} = f_x a + f_y b$$

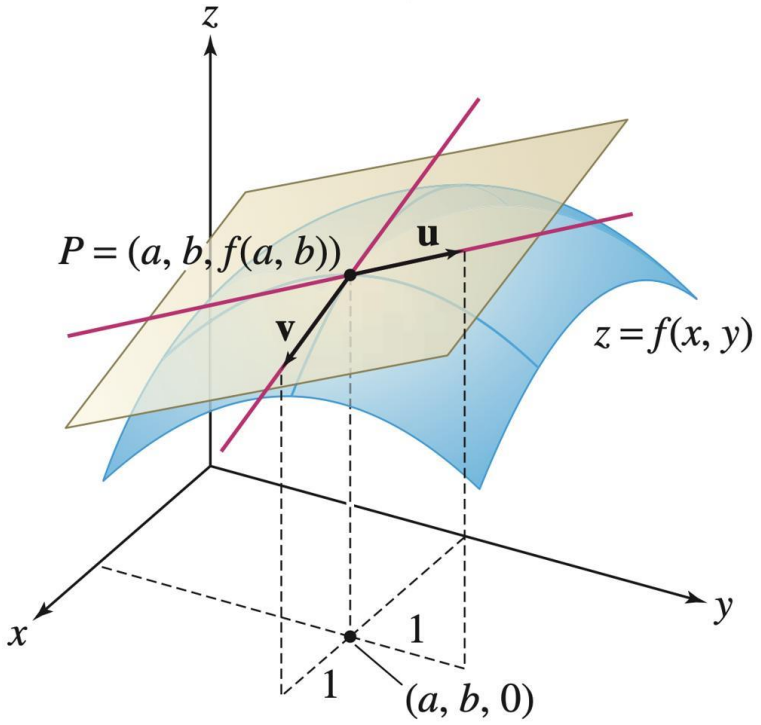
Gradient vs Directional Derivatives

Suppose temperature varies across a metal plate. **The gradient tells** the **direction of greatest heating**. On the other hand,

The directional derivative tells: "If I move in this particular direction, how quickly does temperature change?"

Tangent Plane

A **tangent plane** is a flat plane that **just touches a surface at a particular point** and, near that point, gives the best linear approximation to the surface.



$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Consider the sphere $x^2 + y^2 + z^2 = 25$. Find the equation of **the tangent plane** to the point (0, 3, 4) at the sphere.

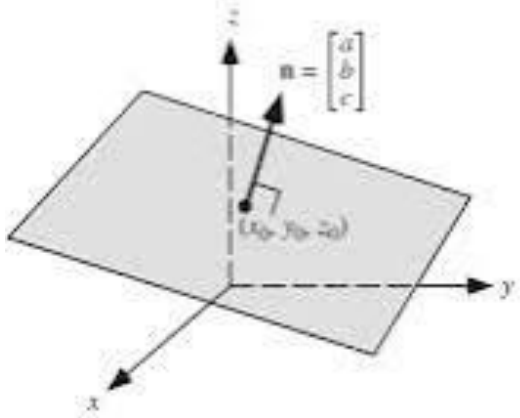
Normal Vector

A **normal vector** to a curve, surface, or plane is a vector that is **perpendicular** to it at a particular point.

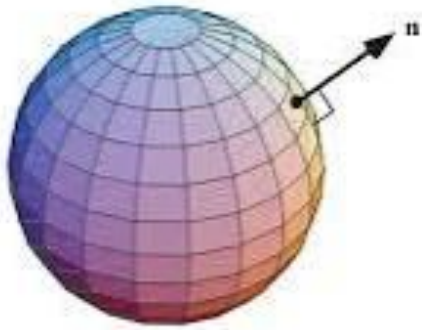
For a surface, a normal vector tells us the **direction in which the surface** is facing.

Example: **For a plane**

Consider $2x + 3y + 4z = 12$

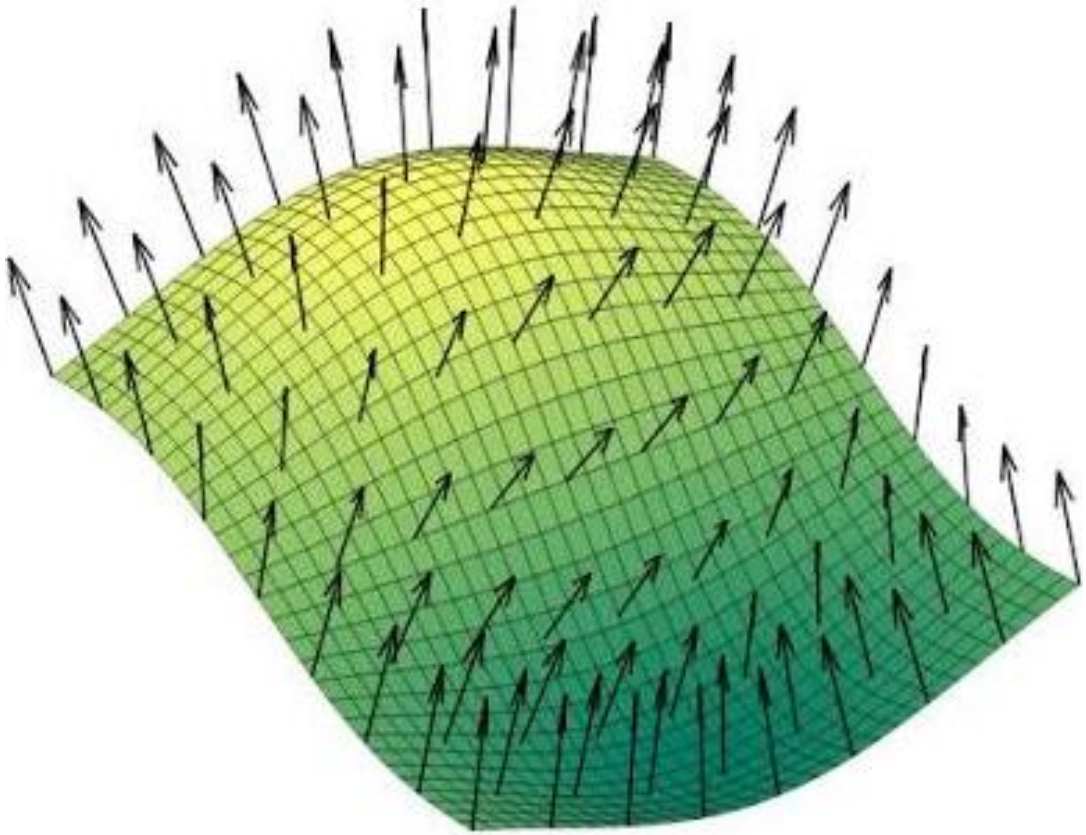


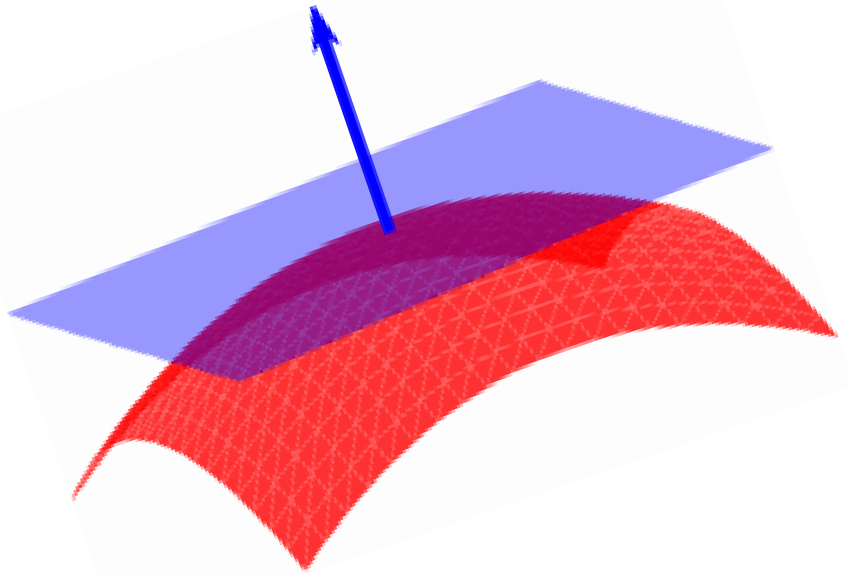
For a sphere $x^2 + y^2 + z^2 = 25.$



Normal Vector

For a surface, a normal vector tells us the direction in which the surface is facing. At different points the **surface has different normal vectors** and is equal to the **normal vector of the tangent plane at this point.**





Normal Vector

Suppose a surface is defined implicitly by $F(x, y, z) = C.$

Then the gradient at a point $P(x_0, y_0, z_0)$, gives the normal vector and represent by $\nabla F(x_0, y_0, z_0)$

Where, $\nabla F = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z}\right)$

Example

$x^2 + y^2 + z^2 = 25.$

Find the normal vector at (3, 4,0)

$\nabla F = (2x, 2y, 2z)$

at (3, 4,0)

$\nabla F = (6, 8, 0)$

The normal vector at (3, 4,0), **n = 6i+8j**

Lagrange Multiplier

Lagrange multipliers are a calculus method used to find the maximum and minimum values of a function when the input values have a strict equality constraint.

The Core Idea

- **Objective function:** The main function $f(x, y)$ that you want to maximize or minimize.
- **Constraint:** A side condition written as $g(x, y) = k$ that your variables must follow.
- **Lagrange multiplier:** A new variable, called λ ***(lambda),*** used to link the function and the constraint

Geometric Meaning

- The highest or lowest point happens where a contour line of your function just touches the constraint line.
- At this touch point, the gradient vector of the function (∇f) points in the exact same or opposite direction as the gradient of the constraint (∇g).
- This means the two gradients are parallel: $\nabla f = \lambda \nabla g$

Example

What is the minimum value of the sum of the square of two numbers when their sum is 4.

Minimize $f(x, y) = x^2 + y^2$ Subject to the constraint $x + y = 4$

Using ordinary calculus

$$y = 4 - x.$$

$$f(x) = x^2 + (4 - x)^2.$$

$$f(x) = 2x^2 - 8x + 16.$$

$$f'(x) = 4x - 8.$$

$$f'(x) = 0$$

$$x = 2.$$

$$y = 4 - 2 = 2.$$

$$\boxed{(2, 2)}$$

Using Lagrange multipliers

$$f(x, y) = x^2 + y^2$$

$$g(x, y) = x + y = 4.$$

$$\nabla f = (2x, 2y)$$

$$\nabla g = (1, 1)$$

Lagrange's condition is

$$(2x, 2y) = \lambda(1, 1)$$

$$2x = \lambda \qquad 2y = \lambda \qquad x = y = 2.$$

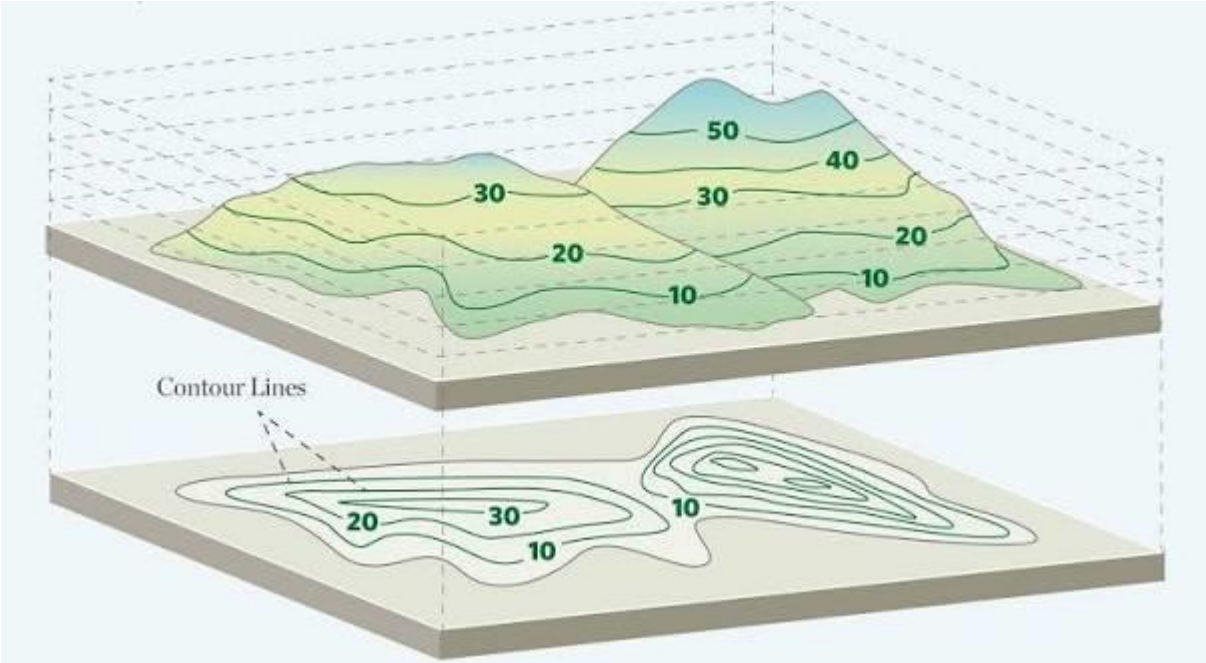
Contour

A **contour** is a curve on which a scalar function has a **constant value**.

Suppose $z = f(x, y)$. If we fix $z = c$, then $\boxed{f(x, y) = c}$, represents a **contour** or **level curve** of f.

In simple words:

A contour joins points where the function **has the same value**.



Scalar Field

A **scalar field** assigns a single number (a scalar) to every point in a region of space. **Definition** In two dimensions: $f(x, y)$. In three dimensions: $f(x, y, z)$

So, a scalar field has the form $\boxed{f: \mathbb{R}^n \to \mathbb{R}}$

Example 1: Temperature

Suppose the temperature in a room is described by $T(x, y, z) = 20 + x^2 + y^2 + z^2$

At the point $(1, 2, 1)$, $T(1, 2, 1) = 20 + 1^2 + 2^2 + 1^2 = 26$

Thus, the temperature at that point is **26°C**. The temperature changes from point to point, but at every point it has only **one value**. Therefore, temperature is a scalar field.

Vector Field

A **vector field** assigns a vector to every point in a region.

Unlike a scalar field, a vector field gives both **magnitude and direction** at every point.

In two dimensions, $\boxed{\mathbf{F}(x,y) = P(x,y)\mathbf{i} + Q(x,y)\mathbf{j}}$

and in three dimensions, $\boxed{\mathbf{F}(x,y,z) = P(x,y,z)\mathbf{i} + Q(x,y,z)\mathbf{j} + R(x,y,z)\mathbf{k}}$

Problem 1

Given $\mathbf{F}(x,y,z) = (x^2, yz, zx)$, find the vector field at P = (2,1,3).

$$\boxed{\mathbf{F}(2,1,3) = (4,3,6)}$$

$$\boxed{\mathbf{F}(x,y,z) = \mathbf{4i} + \mathbf{3j} + \mathbf{6k}}$$

Vector field **F** and a scalar potential function ϕ.

The equation $\boxed{\mathbf{F} = \nabla\phi}$ expresses a very important relationship between a **vector field** F and a **scalar potential function** ϕ.

What does $\nabla\phi$ mean ?

The gradient operator is $\nabla = \frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}$.

Therefore, if $\phi = \phi(x, y, z)$, then $\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$.

So when we write, $\boxed{\mathbf{F} = \nabla\phi}$, we are saying

$$\boxed{\mathbf{F} = \phi_x\mathbf{i} + \phi_y\mathbf{j} + \phi_z\mathbf{k}}$$

Vector field and gradient

Suppose $\phi(x, y) = x^2 + y^2$ is a scalar potential function

Then $\nabla\phi = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}\right) = (2x, 2y)$

Therefore, $\boxed{\mathbf{F} = \nabla\phi = (2x, 2y) = \boldsymbol{2xi + 2yj}}$ is a vector field.

At every point, the vector points **radially outward** from the origin

Conservative vector field

Let $\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$. We say $\mathbf{F}$ is **conservative** vector field if there exists a scalar function $\phi(x, y, z)$ such that $\boxed{\mathbf{F} = \nabla\phi}$

That means $\boxed{P = \frac{\partial \phi}{\partial x}, Q = \frac{\partial \phi}{\partial y}, R = \frac{\partial \phi}{\partial z}}$

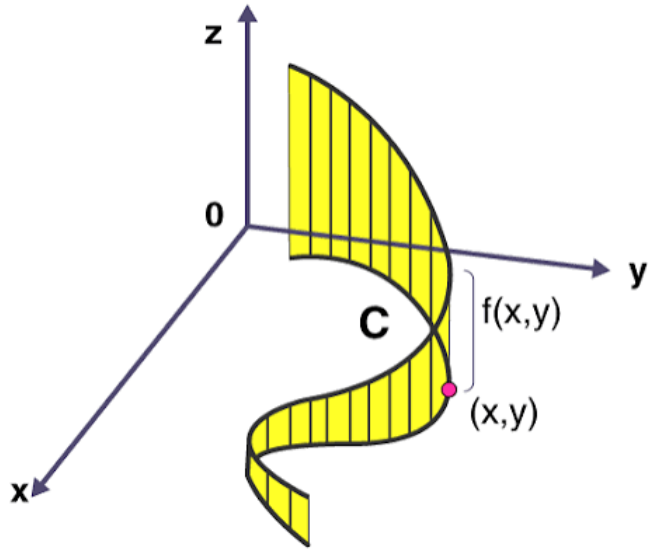
The function $\phi$is called the **potential function**.

So, in simple language: **A conservative vector field is a vector field that can be obtained as the gradient of some scalar potential function**

A vector field is called **conservative** when its line integral (work) between two points depends **only on the initial and final points, not on the path taken**.

Line Integral

A **line integral** is an integral evaluated along a curve or path rather than over an interval on the x axis.. There are two important types:



In ordinary calculus, $\int_a^b f(x)\,dx$, we accumulate the values of $f(x)$ as x moves from a to b.

But suppose instead that we move along a curved path C in the xy-plane:

$$C\colon \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}, a \le t \le b.$$

Then we can accumulate quantities **along that path**. **This gives a line integral.**

I. **Line integral of a scalar field**

II. **Line integral of a vector field**

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Line integral of a scalar field

Suppose we have a **scalar field** $f(x, y)$ and a curve C in the xy-plane. The scalar line integral is

$$\int_C f(x, y)\, ds$$

Parameterize the curve

$$x = x(t), \qquad y = y(t)$$

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$\int_C f(x, y)\, ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

If the curve is given explicitly as

$$y = g(x), \qquad a \le x \le b$$

$$ds = \sqrt{dx^2 + dy^2}$$

$$= \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

$$\int_C f(x, y)\, ds = \int_a^b f(x, g(x)) \sqrt{1 + [g'(x)]^2}\, dx$$

Line Integral of a Scalar Field

Example

Evaluate $\int_C (x+y)\,ds$ where C is the straight-line segment from A(0,0) to B(1,1).

Parameterize the curve

The straight line from $(0, 0)$ to $(1, 1)$ can be written as $x = t,\ \ y = t,\ \ 0 \le t \le 1.$

Thus, $r(t) = t\,\mathrm{i} + t\,\mathrm{j}.$

Find ds

For a plane curve, $ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt.$ Since $\frac{dx}{dt} = 1, \frac{dy}{dt} = 1,$

we get $ds = \sqrt{1^2 + 1^2}\,dt = \sqrt{2}\,dt.$

Therefore, $\int_C (x+y)\,ds = \int_0^1 (2t)(\sqrt{2})\,dt.$

$$= 2\sqrt{2}\int_0^1 t\,dt$$

$$= 2\sqrt{2}\left[\frac{t^2}{2}\right]_0^1$$

$$= \sqrt{2}.$$

Line Integral of a Scalar Field

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Examples

1. **Evaluate** $\boxed{\int_C (x+y)\,ds}$ **, where** $y = x, 0 \le x \le 2$

2. **Evaluate** $\boxed{\int_C (x+y)\,ds}$ **where** $x = y, \quad 0 \le y \le 2.$

3 **Evaluate** $\boxed{\int_C x\ ds}$ **where** $x = t, y = t, 0 \le t \le 1.$

4 **Evaluate** $\boxed{\int_C (x+y)\,ds}$ **where** $x = 2t, y = t, 0 \le t \le 1.$

Line Integral of a Vector Field

$$\mathbf{F} = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}.$$

$$d\mathbf{r} = dx\,\mathbf{i} + dy\,\mathbf{j},$$

$$\mathbf{F} \cdot d\mathbf{r} = P\,dx + Q\,dy.$$

$$\boxed{\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\,dx + Q\,dy}$$

Evaluate $\boxed{\int_C \mathbf{F} \cdot d\mathbf{r}}$ **where** $\mathbf{F} = (y, x) = y\boldsymbol{i} + x\boldsymbol{j}$

And $C: y = x^2, 0 \le x \le 1.$

Since $\mathbf{F} = (y, x)$, **we have** $P = y, Q = x.$

Therefore, $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C y\ dx + x\,dy.$

Since $y = x^2 \quad dy = 2x\,dx.$

Therefore, $\int_C y\ dx + x\,dy = \int_0^1 [x^2\,dx + x(2x\,dx)]\,.$

Thus,

$$= \int_0^1 (3x^2)\,dx$$

$$= 3\left[\frac{x^3}{3}\right]_0^1 = 1$$

Line Integral of a Vector Field

$$\mathbf{F}(x, y, z)$$

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}, \qquad a \le t \le b.$$

$$d\mathbf{r} = \mathbf{r}'(t)\, dt.$$

$$\boxed{\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt}$$

$$\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}.$$

$$x = 2t, \qquad y = t,$$

$$\mathbf{F}(\mathbf{r}(t)) = t\mathbf{i} + 2t\mathbf{j}.$$

$$\boxed{\mathbf{F}(\mathbf{r}(t)) = t\mathbf{i} + 2t\mathbf{j}}.$$

$$\mathbf{F} \cdot d\mathbf{r} = (t\mathbf{i} + 2t\mathbf{j}) \cdot (2\mathbf{i} + \mathbf{j})\, dt.$$

$$\mathbf{i} \cdot \mathbf{i} = 1, \qquad \mathbf{j} \cdot \mathbf{j} = 1, \qquad \mathbf{i} \cdot \mathbf{j} = 0,$$

$$\begin{aligned} \mathbf{F} \cdot d\mathbf{r} &= (2t + 2t)\, dt \\ &= 4t\, dt. \end{aligned}$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 4t\, dt.$$

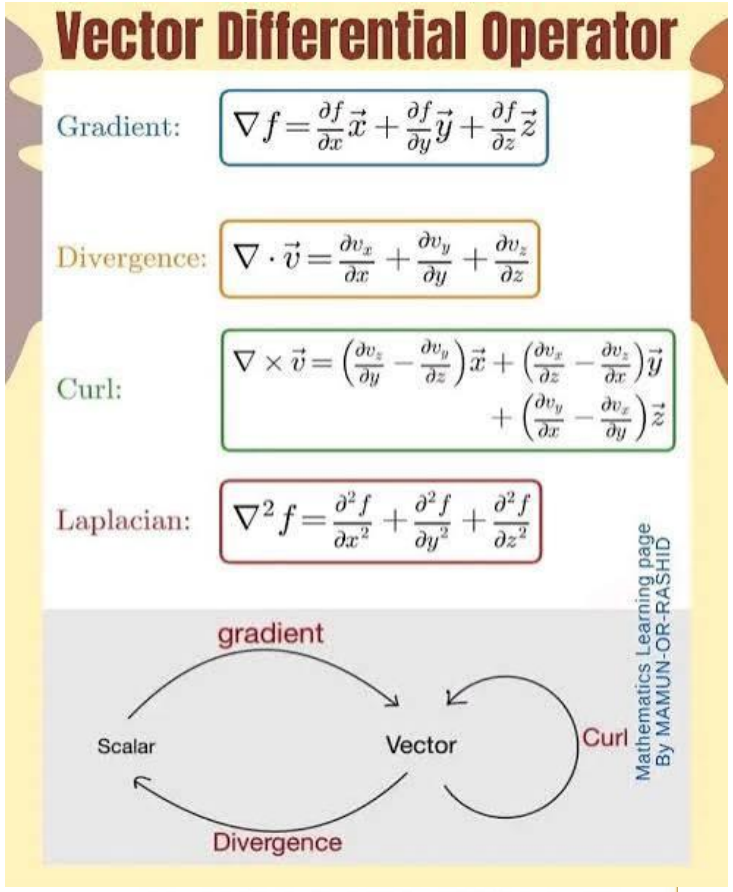
$$\begin{aligned} &= 4\left[\frac{t^2}{2}\right]_0^1 \\ &= 2. \end{aligned}$$

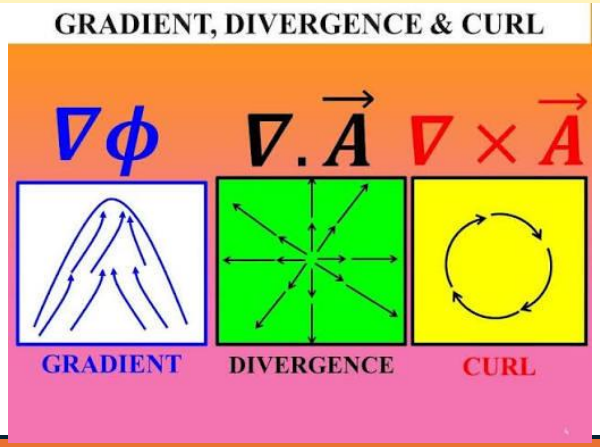
Gradient, Divergence, and Curl

Gradient, Divergence, and Curl These three ideas are fundamental concepts in **vector calculus**. They help us understand how a quantity changes in space and are especially important in **physics, engineering, fluid mechanics, electromagnetism, meteorology, and computer graphics**.

A useful way to remember them is:

- **Gradient** → Which direction does a scalar quantity increase most rapidly?
- **Divergence** → Is a vector field spreading out or converging in?
- **Curl** → Is a vector field rotating around a point?





Divergence

Now we move from scalar fields to **vector fields**.

Suppose $\vec{F} = F_x\hat{\imath} + F_y\hat{\jmath} + F_z\hat{k}$. The divergence is written $\boxed{\nabla \cdot \vec{F}}$ and defined as

$$\boxed{\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}}$$

Notice that divergence uses a **dot product**. That's why it produces a **scalar**.

Gradient, Divergence, and Curl

Positive divergence $\nabla \cdot \vec{F} > 0$. The field behaves like a **source**.

Negative divergence $\nabla \cdot \vec{F} < 0$. The field behaves like a **sink**.

Zero divergence $\nabla \cdot \vec{F} = 0$. There is no net expansion or contraction.

Gradient, Divergence, and Curl

Curl is an operation performed on a **vector field**. For $\vec{F} = F_x\hat{\imath} + F_y\hat{\jmath} + F_z\hat{k}$, the curl is $\boxed{\nabla \times \vec{F}}$

It is defined as

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$\boxed{\nabla \times \mathbf{F} = \mathbf{0}}$

The field is **irrotational**.

Expanding,

$$\nabla \times \vec{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\hat{\imath} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\hat{\jmath} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\hat{k}$$

Unlike divergence, **curl produces a vector**.

Gradient, Divergence, and Curl

Question 1 Find the divergence of $\mathbf{F} = x^2\mathbf{i} + 3xy\mathbf{j} + 2z^2\mathbf{k}$.

$$\nabla \cdot \mathbf{F} = \frac{\partial x^2}{\partial x} + \frac{\partial (3xy)}{\partial y} + \frac{\partial (2z^2)}{\partial z}.$$

Question 2 Find the curl of $\mathbf{F} = (yz)\mathbf{i} + (xz)\mathbf{j} + (xy)\mathbf{k}$.

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Question3

A velocity field of a fluid is given by

$$\mathbf{V} = 2x\mathbf{i} + 3y\mathbf{j} + 4z\mathbf{k}.$$

Determine:

1. The divergence.
2. Whether the fluid is expanding or compressing.
3. Whether the flow is incompressible.

For Compressible Fluid $\nabla \cdot \mathbf{V} \neq 0$

For incompressible Fluid $\nabla \cdot \mathbf{V} = 0$

Gradient, Divergence, and Curl

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$$\boxed{\nabla \times (\nabla f) = 0}$$

In words : **The curl of a gradient is always zero.**

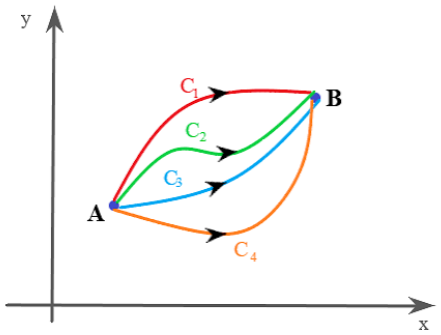
$$\boxed{\nabla \cdot \left(\nabla \times \vec{F}\right) = 0}$$

In words: **The divergence of a curl is always zero.**

Keep this picture in your mind

Mathematical operation	Think of
∇f	Which way is uphill?
$\nabla \cdot \vec{F}$	Is stuff spreading out or gathering?
$\nabla \times \vec{F}$	Is stuff rotating?

Path Independence, Conservative Vector Fields, and Potential Functions



These three ideas are closely connected in vector calculus:

Conservative field ⟺ **Potential function** ⟹ **Path independence**

Path independence:

A line integral is path independent when its value depends only on the start and end points, not on the specific curve taken between them.

A continuous vector field ***F*** has path-independent line integrals in an open, connected region if and only if it meets these equivalent conditions:

- **Conservative Field:** **F** is a conservative vector field.
- **Gradient Field:** **F** is the gradient of a scalar potential function f ($\mathbf{F} = \nabla f$).
- **Closed Loop Integral:** The line integral around any simple closed curve C in the domain is zero ($\oint \mathbf{F} \cdot d\mathbf{r} = 0$).

Path Independence, Conservative Vector Fields, and Potential Functions

A conservative vector field $\vec{F}$ can be expressed as the gradient of a scalar potential function φ, meaning $F=\nabla\varphi$ ***i.e.***

A **vector field *F*** is called **conservative** if it is the gradient of some scalar **potential function** φ.

For a vector field

$$\mathbf{F} = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j},$$

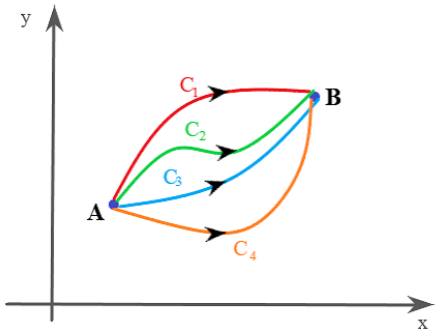
the field is **conservative** in a suitable simply connected domain if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

If $\mathbf{F} = \nabla\phi$, then $\phi(x, y)$ is called a **potential function**.

For a conservative field, the line integral is **path independent**.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A)$$



Path Independence, Conservative Vector Fields, and Potential Functions

A **vector field *F*** is called **conservative** if it is the gradient of some scalar **potential function** φ.

Example

Determine whether the vector field $\mathbf{F} = (2x + 3y)\mathbf{i} + (3x + 4y)\mathbf{j}$ is conservative.
If it is conservative, find a potential function.

$$P = 2x + 3y, \qquad Q = 3x + 4y.$$

$$\frac{\partial P}{\partial y} = 3 \qquad \frac{\partial Q}{\partial x} = 3.$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

Therefore, the field is **conservative**.

Now to find ϕ

$$\frac{\partial \phi}{\partial x} = 2x + 3y.$$

$$\phi = x^2 + 3xy + g(y).$$

$$\frac{\partial \phi}{\partial y} = 3x + g'(y).$$

$$\frac{\partial \phi}{\partial y} = 3x + 4y.$$

$$g'(y) = 4y$$

$$g(y) = 2y^2$$

$$\boxed{\phi = x^2 + 3xy + 2y^2 + C}.$$

Path Independence, Conservative Vector Fields, and Potential Functions

UNITED INTERNATIONAL UNIVERSITY

Q1 Determine whether $\mathbf{F} = (y + 2x)\mathbf{i} + (x + 3y^2)\mathbf{j}$ is conservative.

Q2 Consider $\mathbf{F} = (2xy + 1)\mathbf{i} + x^2\mathbf{j}$. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ $A = (0, 0)$ to $B = (1, 2)$

Q3 Find the potential function for $\mathbf{F} = (6x^2y + 2y^2)\mathbf{i} + (2x^3 + 4xy + 5)\mathbf{j}$.

$$\mathbf{F} = (3x^2 - 6xy + 2)\mathbf{i} + (-3x^2 + 4y)\mathbf{j}.$$

1. Determine whether $\mathbf{F}$ is conservative.
2. Find its potential function.
3. Evaluate the line integral from $A = (1, 1)$ to $B = (2, 3)$.